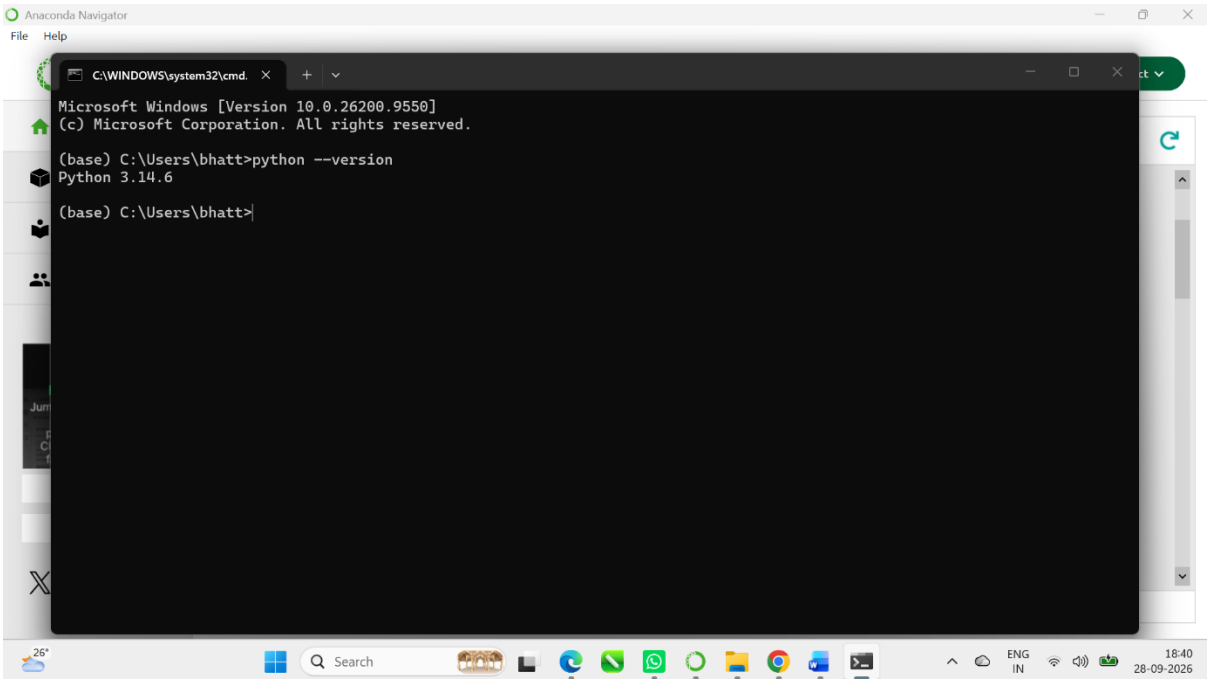
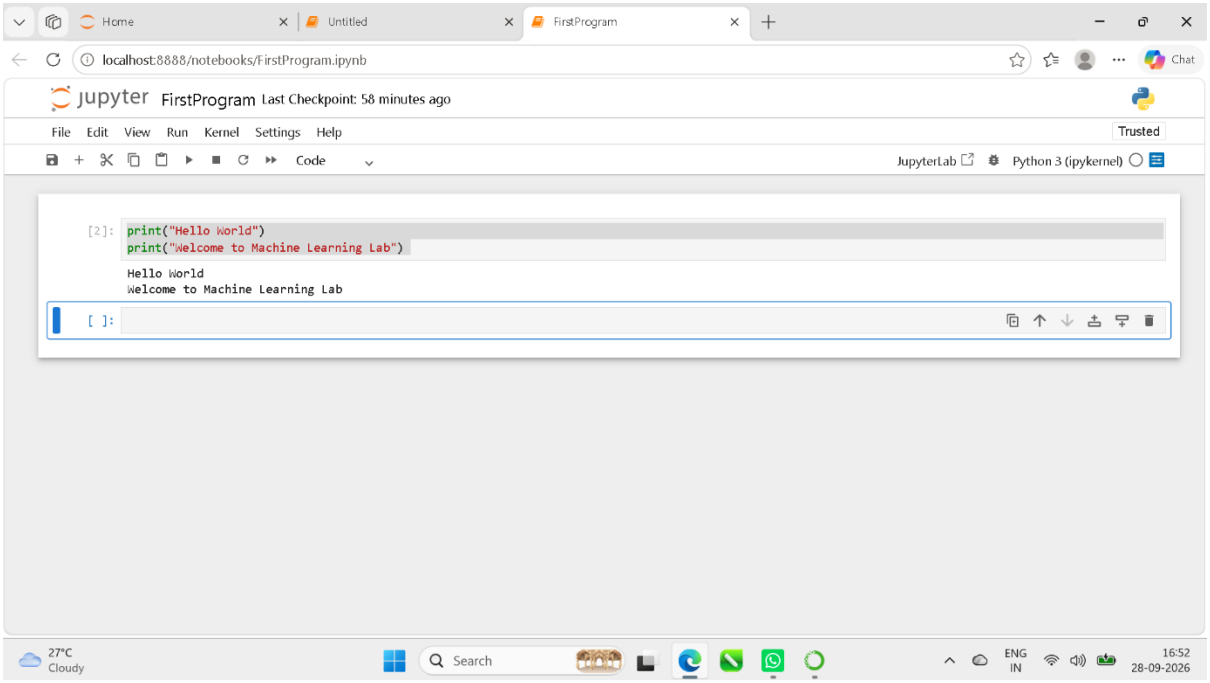
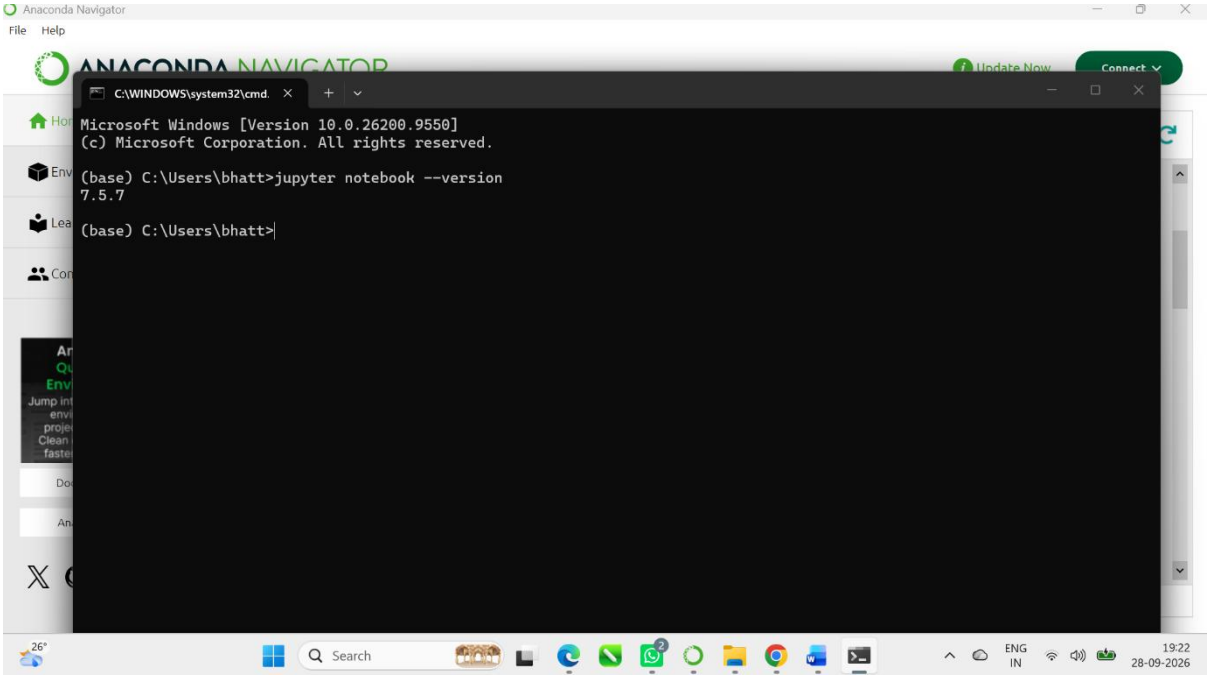
Labsheet 1--- 1 to 10

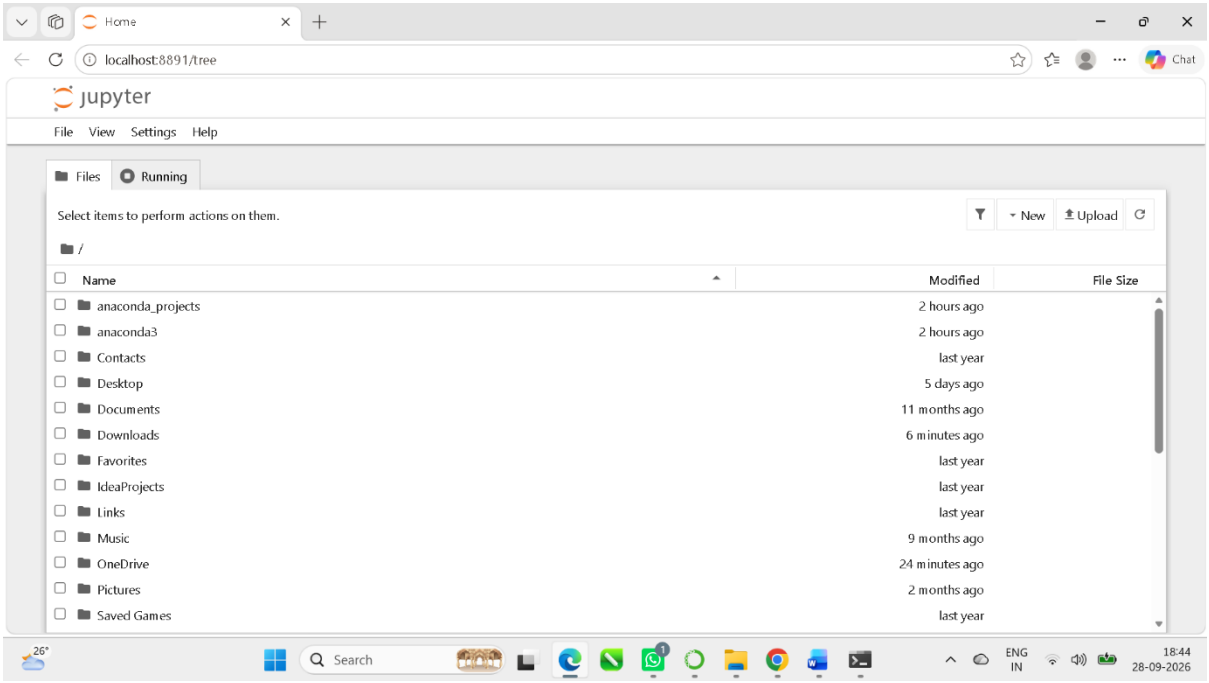
1. python



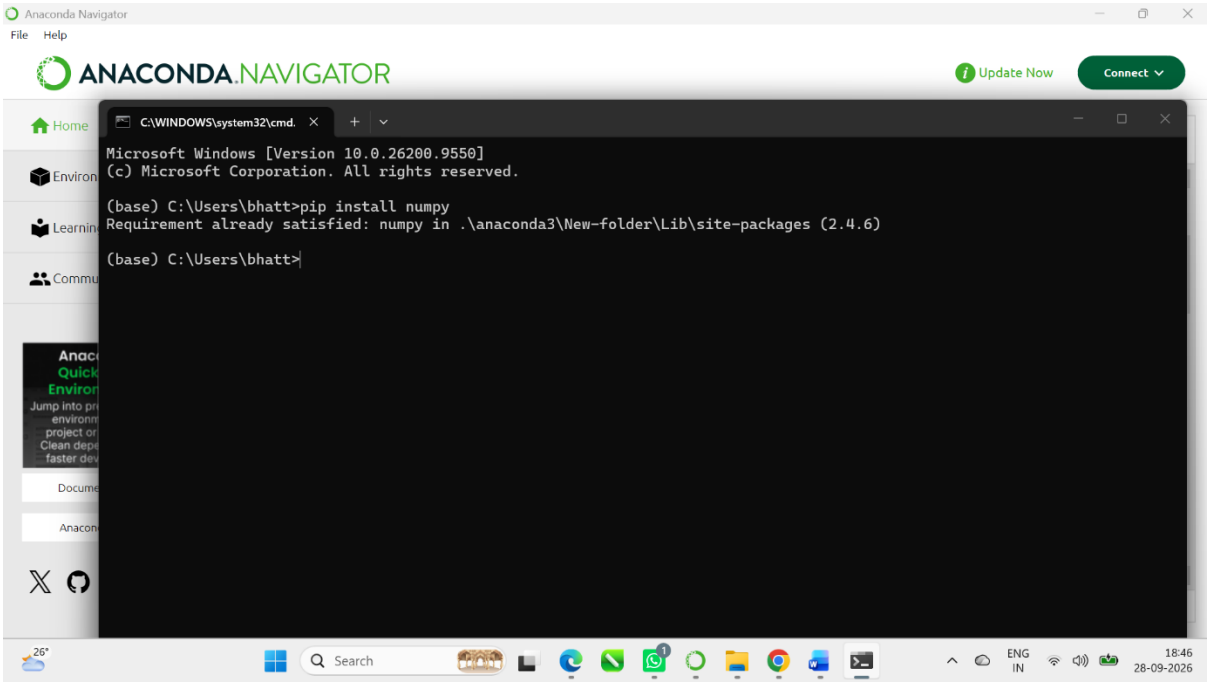


2. jupyter

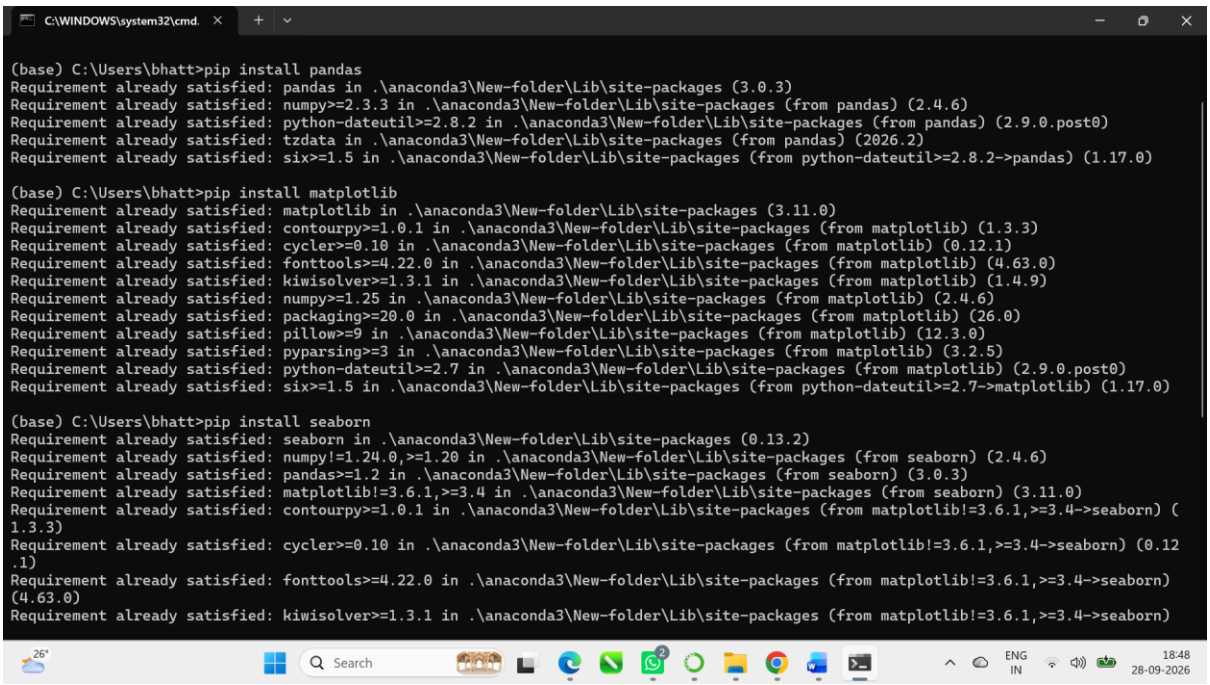

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(c) Microsoft Corporation. All rights reserved.
(base) C:\Users\bhatt>jupyter notebook --version
7.5.7
(base) C:\Users\bhatt>
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anaconda3 2 hours ago
Contacts last year
Desktop 5 days ago
Documents 11 months ago
Downloads 6 minutes ago
Favorites last year
IdeaProjects last year
Links last year
Music 9 months ago
OneDrive 24 minutes ago
Pictures 2 months ago
Saved Games last year
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3. numpy



4. pandas 5. Matplotlib 6. Seaborn 7. Scikit-learn, 8. Tensorflow 9.pytorch 10. gpu check

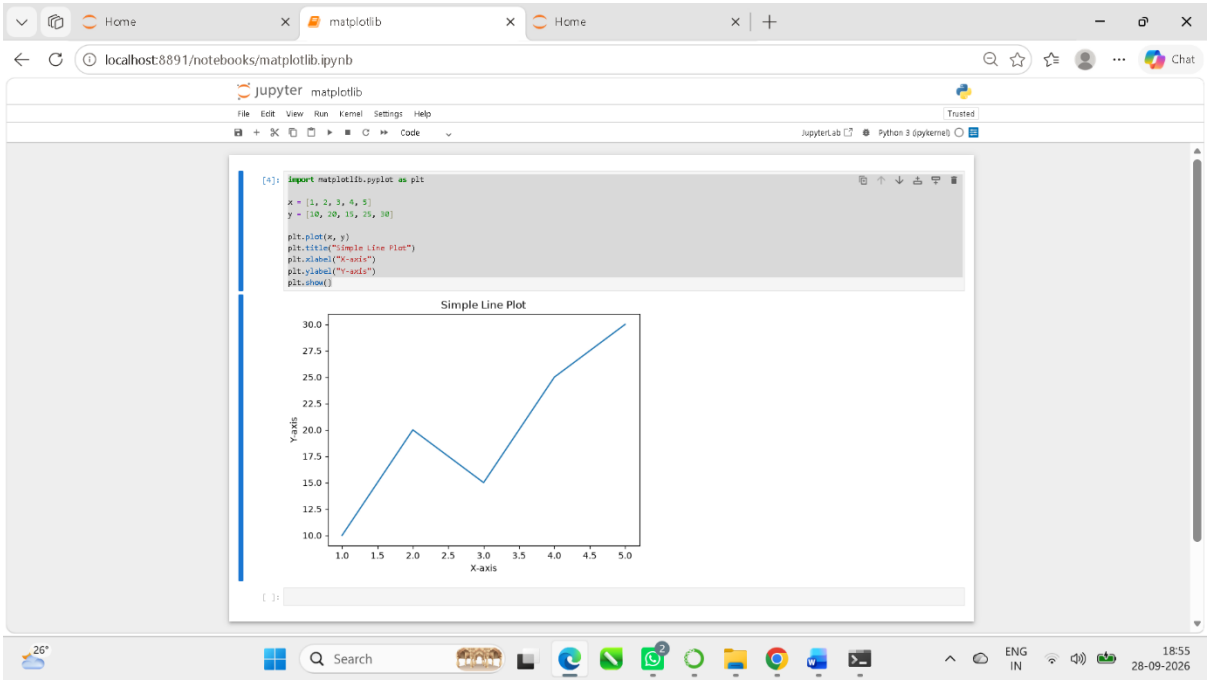


```
C:\WINDOWS\system32\cmd.

(base) C:\Users\bhatt>pip install seaborn
Requirement already satisfied: seaborn in .\anaconda3\New-folder\Lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in .\anaconda3\New-folder\Lib\site-packages (from seaborn) (2.4.6)
Requirement already satisfied: pandas>=1.2 in .\anaconda3\New-folder\Lib\site-packages (from seaborn) (3.0.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in .\anaconda3\New-folder\Lib\site-packages (from seaborn) (3.11.0)
Requirement already satisfied: contourpy>=1.0.1 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (
1.3.3)
Requirement already satisfied: cycler>=0.10 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12
.1)
Requirement already satisfied: fonttools>=4.22.0 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(1.4.9)
Requirement already satisfied: packaging>=20.0 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2
6.0)
Requirement already satisfied: pillow>=9 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.3.0)
Requirement already satisfied: pyparsing>=3 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.
5)
Requirement already satisfied: python-dateutil>=2.7 in .\anaconda3\New-folder\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seabor
n) (2.9.0.post0)
Requirement already satisfied: tzdata in .\anaconda3\New-folder\Lib\site-packages (from pandas>=1.2->seaborn) (2026.2)
Requirement already satisfied: six>=1.5 in .\anaconda3\New-folder\Lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3
.4->seaborn) (1.17.0)

(base) C:\Users\bhatt>pip install scikit-learn
Requirement already satisfied: scikit-learn in .\anaconda3\New-folder\Lib\site-packages (1.9.0)
Requirement already satisfied: numpy>=1.24.1 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (2.4.6)
Requirement already satisfied: scipy>=1.10.0 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (1.18.0)
Requirement already satisfied: joblib>=1.4.0 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (2.23.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (3.5.0)

(base) C:\Users\bhatt>
```



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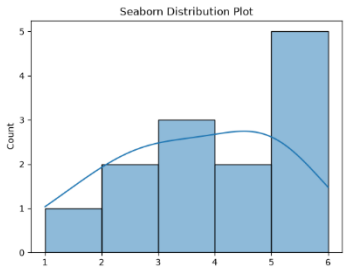
Code

JupyterLab | Python 3 (ipykernel)

```
[1]: import seaborn as sns
     import matplotlib.pyplot as plt

     # sample data
     data = [1,2,2,3,3,3,4,4,5,5,5,5,6]

     sns.histplot(data, kde=True)
     plt.title("Seaborn Distribution Plot")
     plt.show()
```




```
[ ]:
```

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Code

JupyterLab | Python 3 (ipykernel)

```
[1]: import sklearn
     print(sklearn.__version__)
```

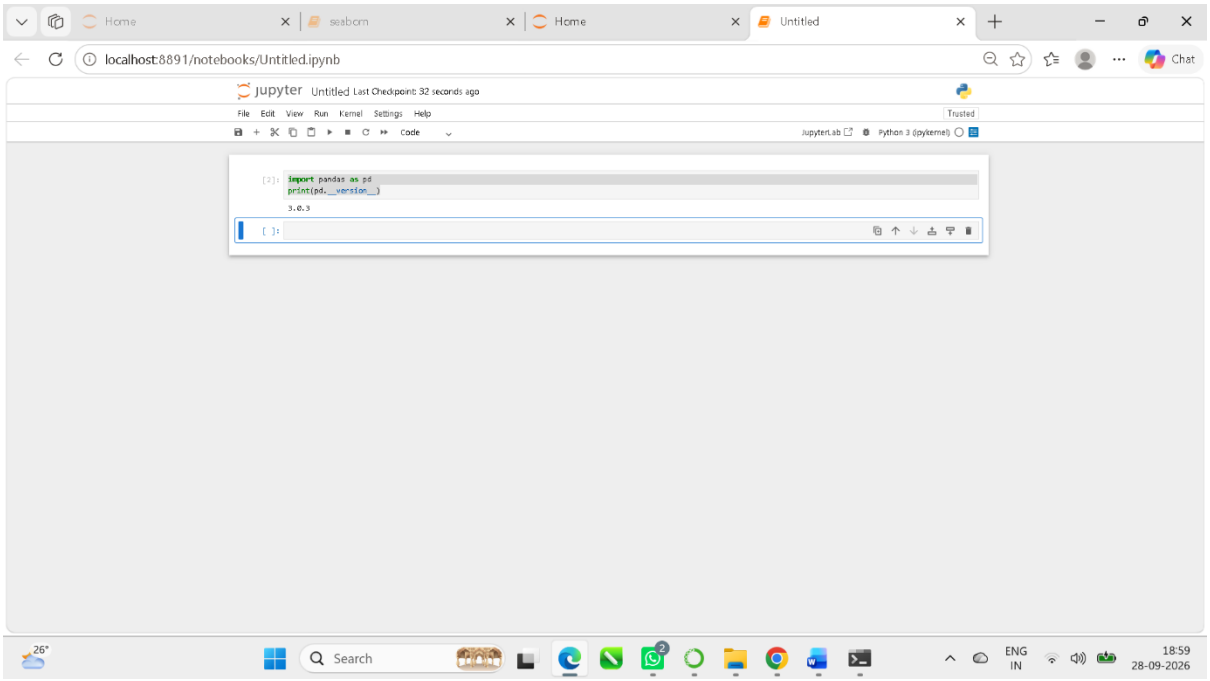
```
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```

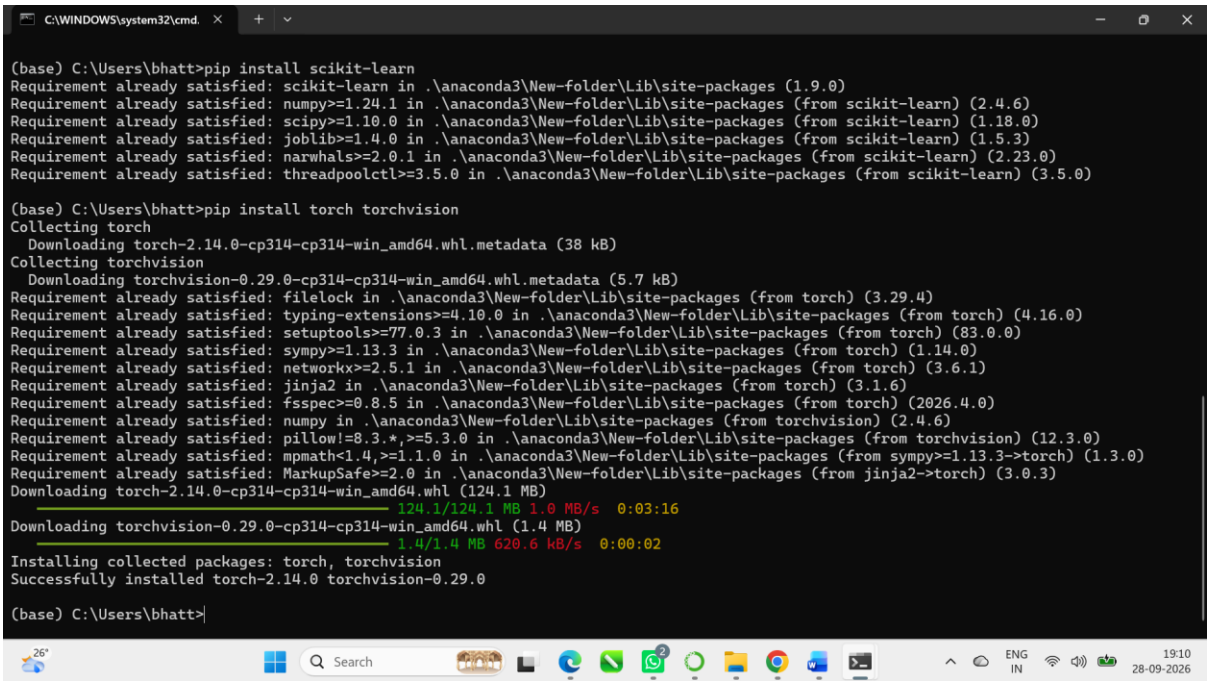
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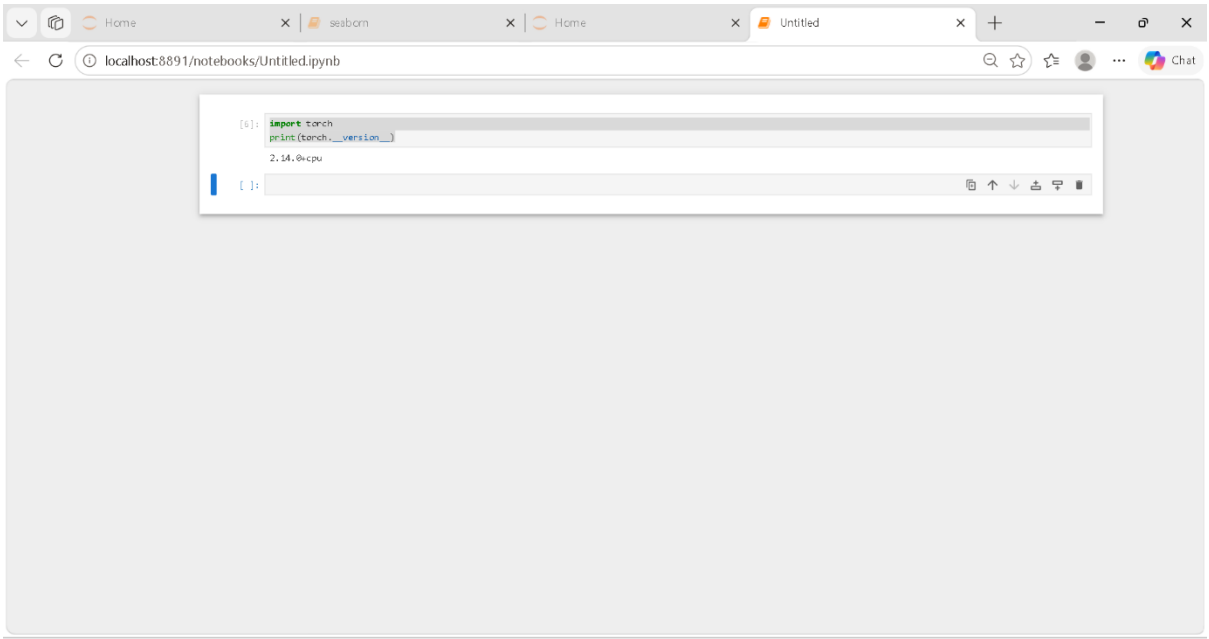
26°

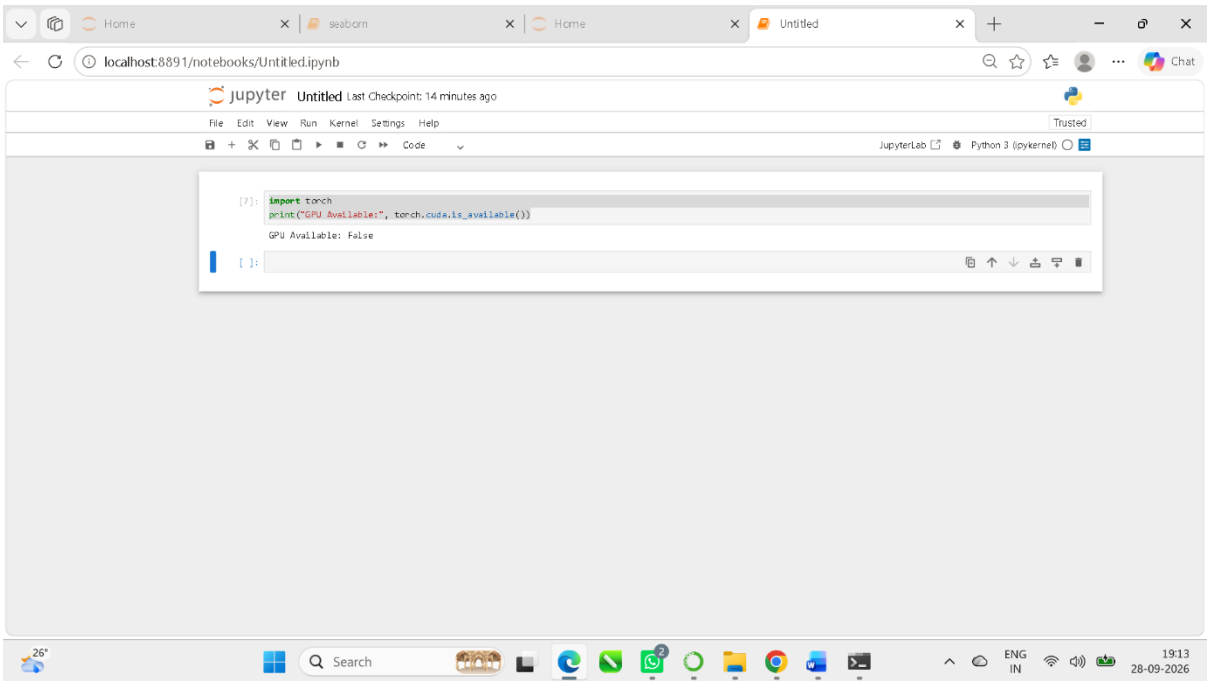
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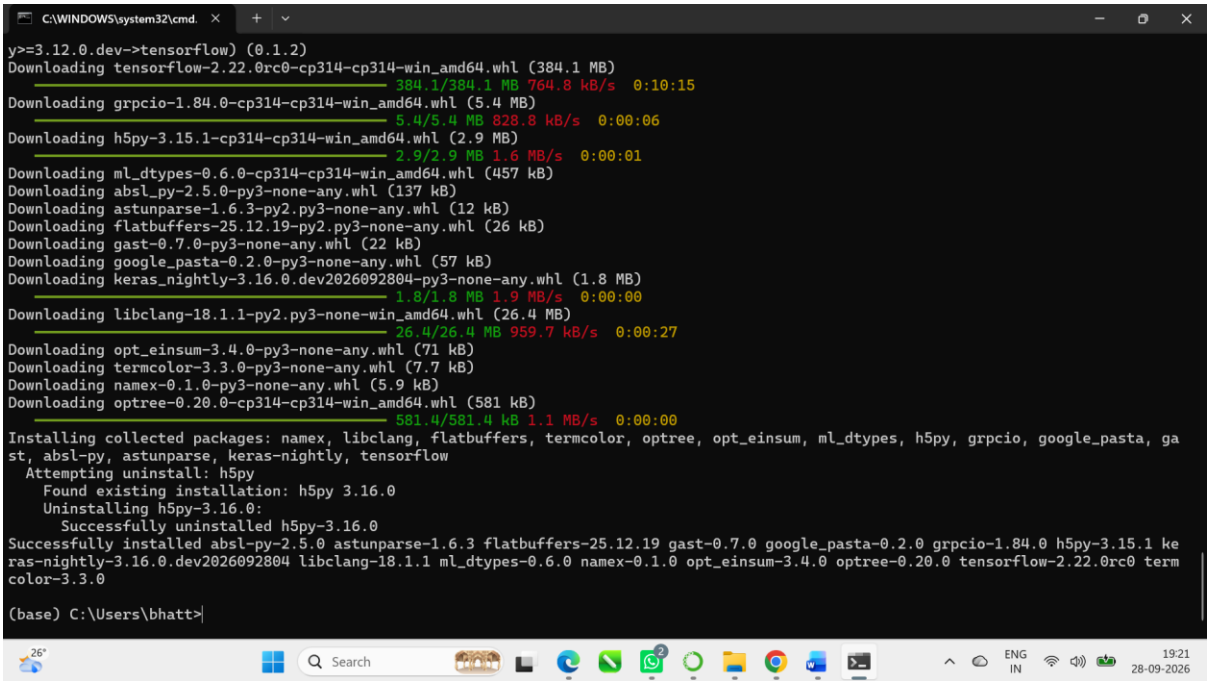
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Python 3 (ipykernel)
[2]: import pandas as pd
print(pd.__version__)
3.0.3
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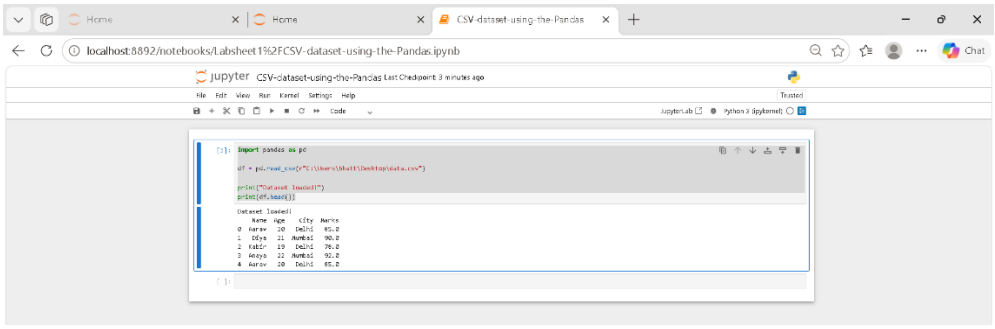

C:\WINDOWS\system32\cmd.
(base) C:\Users\bhatt>pip install scikit-learn
Requirement already satisfied: scikit-learn in .\anaconda3\New-folder\Lib\site-packages (1.9.0)
Requirement already satisfied: numpy>=1.24.1 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (2.4.6)
Requirement already satisfied: scipy>=1.10.0 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (1.18.0)
Requirement already satisfied: joblib>=1.4.0 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (1.5.3)
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Requirement already satisfied: threadpoolctl>=3.5.0 in .\anaconda3\New-folder\Lib\site-packages (from scikit-learn) (3.5.0)
(base) C:\Users\bhatt>pip install torch torchvision
Collecting torch
Downloading torch-2.14.0-cp314-cp314-win_amd64.whl.metadata (38 kB)
Collecting torchvision
Downloading torchvision-0.29.0-cp314-cp314-win_amd64.whl.metadata (5.7 kB)
Requirement already satisfied: filelock in .\anaconda3\New-folder\Lib\site-packages (from torch) (3.29.4)
Requirement already satisfied: typing-extensions>=4.10.0 in .\anaconda3\New-folder\Lib\site-packages (from torch) (4.16.0)
Requirement already satisfied: setuptools>=77.0.3 in .\anaconda3\New-folder\Lib\site-packages (from torch) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in .\anaconda3\New-folder\Lib\site-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in .\anaconda3\New-folder\Lib\site-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in .\anaconda3\New-folder\Lib\site-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in .\anaconda3\New-folder\Lib\site-packages (from torch) (2026.4.0)
Requirement already satisfied: numpy in .\anaconda3\New-folder\Lib\site-packages (from torchvision) (2.4.6)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in .\anaconda3\New-folder\Lib\site-packages (from torchvision) (12.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in .\anaconda3\New-folder\Lib\site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in .\anaconda3\New-folder\Lib\site-packages (from jinja2->torch) (3.0.3)
Downloading torch-2.14.0-cp314-cp314-win_amd64.whl (124.1 MB)
124.1/124.1 MB 1.0 MB/s 0:03:16
Downloading torchvision-0.29.0-cp314-cp314-win_amd64.whl (1.4 MB)
1.4/1.4 MB 620.6 kB/s 0:00:02
Installing collected packages: torch, torchvision
Successfully installed torch-2.14.0 torchvision-0.29.0
(base) C:\Users\bhatt>
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[6]:
import torch
print(torch.__version__)
2.14.0+cpu
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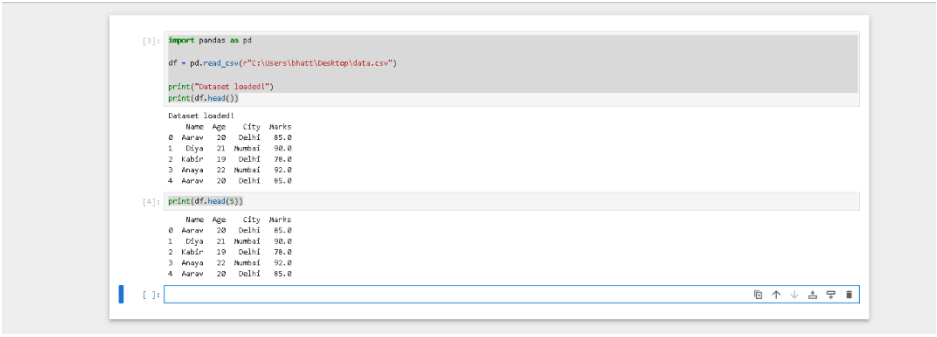

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Python 3 (ipykernel)
[7]:
import torch
print("GPU Available:", torch.cuda.is_available())
GPU Available: False
[ ]:
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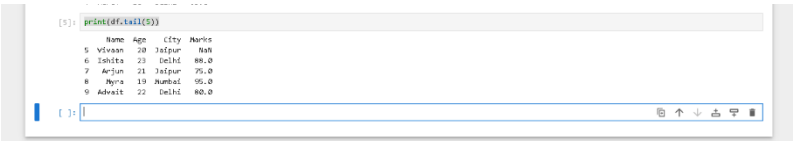
11. Csv



12. print the first 5 rows of the data set



13. Print the last 5 rows of the dataset.

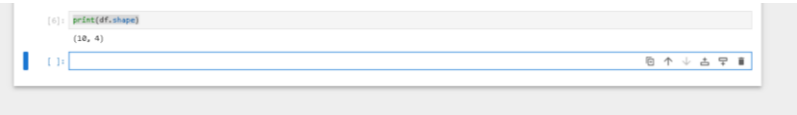


```
[5]: print(df.tail(5))

      Name  Age    City  Marks
5   Vivaan   20  Jaipur    NaN
6   Ishita   23   Delhi   88.0
7   Arjun    21  Jaipur   75.0
8   Myra     19  Mumbai   95.0
9   Advait   22   Delhi   80.0

[ ]:
```

14. Get the number of rows and columns in the dataset.

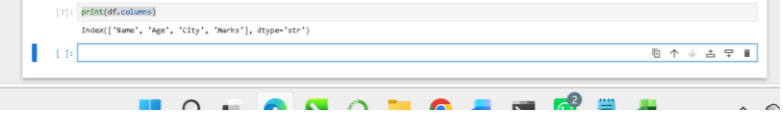


```
[6]: print(df.shape)

(10, 4)

[ ]:
```

15. Display the column names of the dataset.

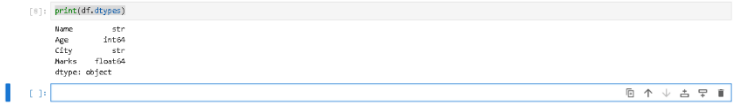


```
[7]: print(df.columns)

Index(['Name', 'Age', 'City', 'Marks'], dtype='str')

[ ]:
```

16. Get the data types of each column.

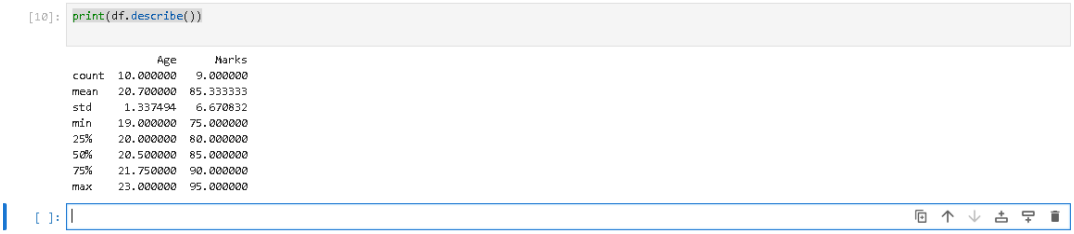


```
[8]: print(df.dtypes)

Name      str
Age     int64
City      str
Marks   float64
dtype: object

[ ]:
```

17. Generate descriptive statistics of numerical features.

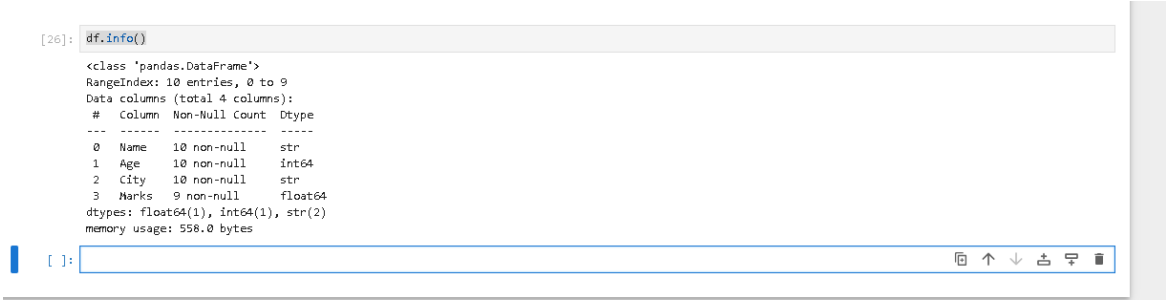


```
[10]: print(df.describe())

             Age      Marks
count  10.000000   9.000000
mean   20.700000  85.333333
std     1.337494   6.670832
min    19.000000  75.000000
25%    20.000000  80.000000
50%    20.500000  85.000000
75%    21.750000  90.000000
max    23.000000  95.000000

[ ]:
```

18. Display complete information about the dataset using info().



```
[26]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Name    10 non-null     str
 1   Age     10 non-null     int64
 2   City    10 non-null     str
 3   Marks   9 non-null      float64
dtypes: float64(1), int64(1), str(2)
memory usage: 558.0 bytes

[ ]:
```

19. Check for missing values in the dataset.

```
[9]: print(df.isnull().sum())
```

```
Name     0
Age      0
City     0
Marks    1
dtype: int64
```

```
[ ]:
```

20. Count the total missing values in each column.

```
[27]: print(df.isnull().sum())
```

```
Name     0
Age      0
City     0
Marks    1
dtype: int64
```

```
[ ]:
```

21. Display unique values present in a selected column.

```
[28]: print(df['City'].unique())
```

```
<ArrowStringArray>
['Delhi', 'Mumbai', 'Jaipur']
Length: 3, dtype: str
```

```
[ ]:
```

22. Count the frequency of each unique value in a column.

```
[29]: print(df['City'].value_counts())
```

```
City
Delhi     5
Mumbai    3
Jaipur    2
Name: count, dtype: int64
```

```
[ ]:
```

23. Rename one or more columns in a dataset.

```
[30]: df = df.rename(columns={'Name': 'Student_Name'})
print(df.columns)
```

```
Index(['Student_Name', 'Age', 'City', 'Marks'], dtype='str')
```

```
[ ]:
```

24. Select specific rows and columns using loc[].

```
[31]: print(df.loc[0:2, ['Student_Name', 'Age']])
```

```
  Student_Name  Age
0        Aarav   20
1         Diya   21
2        Kabir   19
```

```
[ ]:
```

25. Select specific rows and columns using iloc[].

```
[32]: print(df.iloc[0:3, 0:2])
```

```
  Student_Name  Age
0        Aarav   20
1         Diya   21
2        Kabir   19
```

```
[ ]:
```

26. Filter records based on a given condition.

```
[33]: print(df[df['Age'] > 20])
```

```
  Student_Name  Age    City  Marks
1         Diya   21  Mumbai   90.0
3        Anaya   22  Mumbai   92.0
6       Ishita   23   Delhi   88.0
7        Arjun   21  Jaipur   75.0
9       Advait   22   Delhi   80.0
```

```
[ ]:
```

27. Sort the dataset using one or more columns.

```
[34]: df_sorted = df.sort_values(by='Age')
      print(df_sorted)
```

```
  Student_Name  Age    City  Marks
2        Kabir   19   Delhi   78.0
8         Myra   19  Mumbai   95.0
5       Vivaan   20  Jaipur    NaN
0        Aarav   20   Delhi   85.0
4        Aarav   20   Delhi   85.0
7        Arjun   21  Jaipur   75.0
1         Diya   21  Mumbai   90.0
3        Anaya   22  Mumbai   92.0
9       Advait   22   Delhi   80.0
6       Ishita   23   Delhi   88.0
```

```
[ ]:
```

28. Add a new column to an existing dataset.

```
[35]: df['Status'] = 'Pass'
      print(df.head())
```

```
  Student_Name  Age    City  Marks Status
0        Aarav   20   Delhi   85.0   Pass
1         Diya   21  Mumbai   90.0   Pass
2        Kabir   19   Delhi   78.0   Pass
3        Anaya   22  Mumbai   92.0   Pass
4        Aarav   20   Delhi   85.0   Pass
```

```
[ ]:
```

---

29. Delete an existing column from the dataset.

```
[36]: df = df.drop(columns=['Status'])
      print(df.columns)
```

```
Index(['Student_Name', 'Age', 'City', 'Marks'], dtype='str')
```

```
[ ]:
```

30. Read the newly saved CSV file.

```
[37]: df_new = pd.read_csv(r"C:\Users\bhatt\Desktop\cleaned_data.csv")
      print(df_new.head())
```

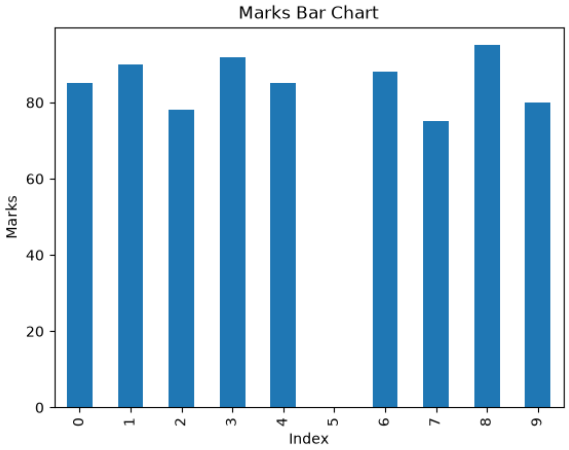
```
     Name  Age    City  Marks
0   Aarav   20   Delhi   85.0
1    Diya   21  Mumbai   90.0
2   Kabir   19   Delhi   78.0
3   Anaya   22  Mumbai   92.0
4  Vivaan   20  Jaipur    NaN
```

```
[ ]:
```

31. Plot a bar chart of a column (e.g., Marks)

```
[25]: import matplotlib.pyplot as plt

df['Marks'].plot(kind='bar')
plt.title('Marks Bar Chart')
plt.xlabel('Index')
plt.ylabel('Marks')
plt.show()
```




```
[ ]:
```

32. Load a dataset directly from the Scikit-learn library.

```
[38]: from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
df_iris = pd.DataFrame(iris.data, columns=iris.feature_names)
df_iris['target'] = iris.target
print(df_iris.head())
```

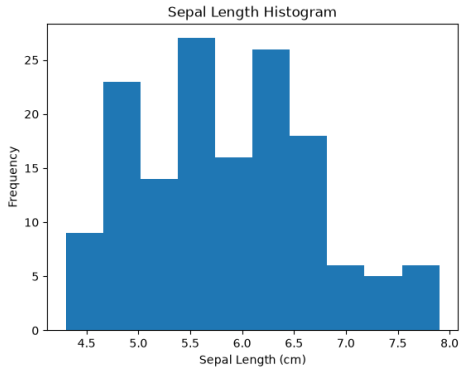
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target
0       0
1       0
2       0
3       0
4       0
```

33. Create a histogram for a numerical feature using Matplotlib.

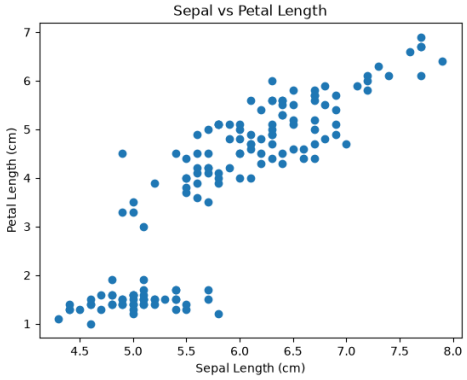
```
[39]: import matplotlib.pyplot as plt

plt.hist(df_iris['sepal length (cm)'], bins=10)
plt.title('Sepal Length Histogram')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Frequency')
plt.show()
```



34. Create a scatter plot to visualize the relationship between two variables.

```
[40]: plt.scatter(df_iris['sepal length (cm)'], df_iris['petal length (cm)'])
plt.title('Sepal vs Petal Length')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Petal Length (cm)')
plt.show()
```




```
[ ]:
```

35. Generate a correlation matrix and visualize it using a heatmap.

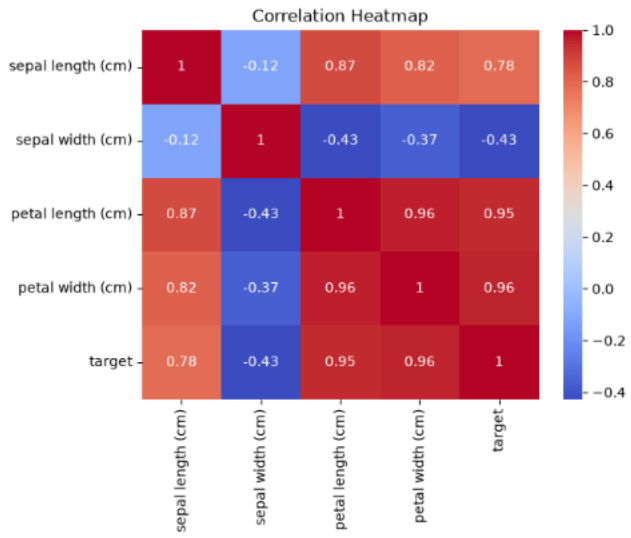
```
[41]: import seaborn as sns

corr = df_iris.corr()
print(corr)

sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

```
                   sepal length (cm)  sepal width (cm)  petal length (cm)  \
sepal length (cm)           1.000000         -0.117570           0.871754
sepal width (cm)           -0.117570          1.000000          -0.428440
petal length (cm)           0.871754         -0.428440           1.000000
petal width (cm)            0.817941         -0.366126           0.962865
target                      0.782561         -0.426658           0.949035

                   petal width (cm)    target
sepal length (cm)          0.817941  0.782561
sepal width (cm)          -0.366126 -0.426658
petal length (cm)          0.962865  0.949035
petal width (cm)           1.000000  0.956547
target                     0.956547  1.000000
```




```
[ ]:
```

Files:

https://drive.google.com/drive/folders/1epBMNEyLBxiErOz8wHkqIveslfs5yEOk?usp=drive_link